DOPPELGANGER
Sound Effects and Their Synthetic Twins

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Abstract

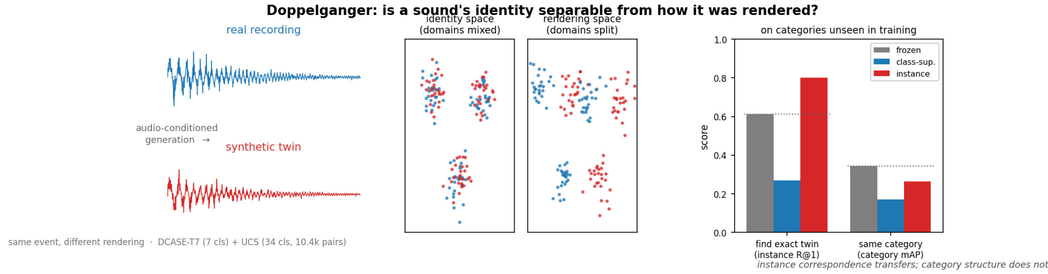
Audio-conditioned generators now produce synthetic sound effects from real recordings, so the real and synthetic versions of an event increasingly coexist in sound libraries and in the corpora used to train audio models—yet no benchmark measures whether a representation can match a synthetic clip to the specific real recording it was generated from. I introduce Doppelganger, a benchmark for matching sound effects across the synthetic–real boundary, pairing $10{,}420$ real clips across 34 everyday sound events each with an audio-conditioned synthetic twin, alongside a controlled 7-class corpus. Off-the-shelf audio encoders do not cross the boundary cleanly. Making the embedding ignore the boundary the standard way—training it on sound-event labels—works on familiar sounds but backfires on new ones, dropping below the untrained encoder. Training on the pairs instead—a clip and its own synthetic twin—generalizes. On sound events held out of training, it recovers the exact real source about 80% of the time (chance 0.03%; 73–99% across six encoders), whereas no objective meaningfully improves category-level recognition on those unseen events. The learned matching is specific to one generator—it survives changes to that generator's settings but not a switch to a different generator, and collapses for the text-only generators tested. A human annotation baseline (49 listeners) lands well above chance but below the models on the same trials. Synthetic twins fool people into calling them real about 29% of the time, yet a generator-specific detector separates these audio-conditioned twins from real recordings perfectly.

1 Introduction

Audio-conditioned generators now turn real sound effects into synthetic variants, so the real and synthetic versions of an event increasingly coexist—side by side in sound libraries, and mixed together in the web-scale corpora used to train audio models. This raises questions the field's usual tools do not answer—questions not about whether two clips share a sound event, but about whether one specific synthetic clip is the counterpart of one specific real recording. Single-domain embedding benchmarks [Turian et al., 2022, Heigold et al., 2026] score whether an embedding groups similar-sounding clips, but treat all audio as one domain. Distributional generative metrics such as Fréchet Audio Distance [Kilgour et al., 2019] ask whether a generator's output distribution resembles real audio, not whether a given output preserved the event it was conditioned on. And synthetic-audio detection [Ouajdi et al., 2024, Yin et al., 2025] asks only real-or-fake, discarding the correspondence between a synthetic clip and its source.

I argue these are facets of one capability—representing the identity of a sound event across the synthetic–real boundary (Figure 1)—and pose it as retrieval:

Figure 1: The synthetic–real matching task and the instance-versus-category comparison on unseen sound events.



Notes. **Left:** each real recording is paired with an audio-conditioned synthetic twin of the same event; I ask whether an embedding can represent the event identity while ignoring how it was rendered. **Middle:** an invariant embedding mixes the two domains within each event cluster, while a sensitive one splits them—the two axes this benchmark separates. **Right:** on sound types never seen during training, an embedding trained to match each clip to its own synthetic twin (red) finds the exact real twin far more often than the untrained baseline (dotted line), while no training meaningfully improves retrieval of other clips of the same kind. Matching a specific sound to its twin transfers to new sound types; recognizing the broad sound event does not.

> Can a representation recognize that a synthetic clip and a real recording are counterparts of the same event, and does that ability transfer to event types not seen during training?

Transfer is the objective. No benchmark can enumerate every sound, so a representation is useful only if the correspondence survives on sound events it never saw. I frame the task as retrieval rather than detection—detection rewards exploiting the synthetic–real gap, whereas retrieval rewards removing it, which is what cross-domain search, real/synthetic deduplication, and clip-by-clip generator evaluation actually need.

This work shows that the correspondence between a clip and its audio-conditioned synthetic twin is a learnable object that transfers across sound events. Sound-event identity is not, and class-supervised invariance—the standard recipe—actually degrades unseen sound events below the untrained baseline. The same machinery yields a complementary sensitive head whose real-vs-synthetic direction separates a generator's outputs from real audio clip-by-clip: a paired complement to FAD that audits what a generator did to a specific input rather than to a distribution. Both capabilities turn out to be generator-specific. The correspondence vanishes for text-only generators, and the sensitive axis rates an unseen generator's clips as nearly real—which bounds both claims, and points at how this benchmark can test and guide generators that condition on audio.

What this is useful for. The released embeddings support (i) cross-domain retrieval—find the real source of a synthetic clip, or search a real library with a generated query; (ii) dataset hygiene—cluster synthetic derivatives with their real sources to catch leakage and near-duplicate contamination before they enter a training set; and (iii) generator evaluation—scoring, one clip at a time, whether a synthetic-sound model preserved the event it was conditioned on, and ranking outputs or systems by it.

Contributions. **(C1)** a two-part benchmark—a controlled 7-class corpus (DCASE-T7, from the Detection and Classification of Acoustic Scenes and Events challenge) and a diverse, instance-paired 34-event corpus—with leakage-safe splits and a reproducible generation recipe. **(C2)** a measured dissociation: an instance-contrastive objective generalizes synthetic–real instance correspondence to unseen sound events (mean full-gallery R@1 0.800, with positive margins in all five folds), whereas class-supervised invariance does not (and degrades unseen sound events below baseline); robust over five folds and six encoders—supervised, audio-text, and self-supervised—five of which did not participate in corpus verification. **(C3)** a complementary sensitive head that separates a generator's outputs from real recordings clip-by-clip (within-generator AUC, area under the ROC curve, 1.0)—and the finding that it shares the instance head's boundary: it neither grades fidelity continuously nor transfers to an unseen generator, so realness detection is per-generator too. **(C4)**

Table 1: Doppelganger compared with related audio benchmarks.

benchmark	domain split	correspondence	synthetic source	evaluation
HEAR / MSEB / MAEB	single-domain	sound event	—	class / retrieval / cluster
EnvSDD / deepfake-audio	synth vs. real	— (exploit gap)	mixed generators	detection (classifier)
FAD	real vs. generated	—	generators	distributional distance
Doppelganger (ours)	synth $\leftrightarrow$ real	instance + sound event	audio-conditioned twin	retrieval (R@1, mAP)

Notes. Rows are prior lines of work; columns are the axes that separate them. domain split: whether the benchmark spans one audio domain or both real and synthetic. correspondence: the unit two clips are matched on—sound event, or the specific instance ("exploit gap" means the method only separates real from synthetic, matching nothing). synthetic source: where synthetic audio, if any, comes from. evaluation: the metric. Doppelganger is the only row that pairs each real recording with an audio-conditioned synthetic twin and evaluates both sound event- and instance-level correspondence across the synthetic–real boundary.

released artifacts—benchmark, recipe, CLAP-verification, trained heads, and a reusable transform that adjusts any new clip's embedding.

2 Related Work

Four lines of work bound this task. Table 1 places them along the axes that distinguish Doppelganger—which domains they span, whether they match at the sound event or the instance level, where their synthetic audio comes from, and how they evaluate—and I discuss each in turn. The short version: embedding benchmarks match within a single domain, detection exploits the synthetic–real gap, and generative metrics score realism distributionally. None pairs each real clip with an audio-conditioned synthetic twin and measures instance-level correspondence across the boundary.

Audio-representation benchmarks (row 1). HEAR [Turian et al., 2022] standardized frozen-embedding evaluation across dozens of audio tasks, and the recent massive benchmarks MSEB [Heigold et al., 2026] and MAEB [El Assadi et al., 2026] scale this to classification, retrieval, and clustering over many datasets. They ask whether an embedding groups semantically similar clips, but treat audio as a single domain. None tests invariance to how a sound was produced, or asks whether a synthetic clip can be matched to the specific real recording it imitates. Embedding-based retrieval and representation repair are more developed in text-as-data settings, from sentence embeddings and dense passage retrieval [Reimers and Gurevych, 2019, Karpukhin et al., 2020] to relevance annotation for retrieval-augmented generation [Ni et al., 2025] and the removal of document-source confounds from embedding similarity and clustering [Fan et al., 2025]. Doppelganger brings this retrieval-and-representation evaluation perspective to audio. It adds exactly the missing axis and evaluates the same frozen encoders these benchmarks rank—CLAP (Wu et al., 2023; Elizalde et al., 2023), PANNs [Kong et al., 2020], AST [Gong et al., 2021], and the SSL families BEATs [Chen et al., 2023], M2D [Niizumi et al., 2024], and AudioMAE [Huang et al., 2022]—under a cross-domain, instance-level protocol.

Synthetic-audio detection (row 2). The closest prior use of the core corpus is deepfake-environmental-audio detection: Ouajdi et al. [2024] train a real-vs-synthetic classifier on the DCASE-T7 Foley set, and EnvSDD [Yin et al., 2025] scales detection to a large benchmark. These pursue the opposite direction—they learn representations that exploit the synthetic–real gap, whereas I ask when that gap can be removed while event identity is kept. I reproduce their objective as a controlled sensitive head and show it is one end of a single axis whose other end (the invariant head) supports cross-domain retrieval. Detection and correspondence are thus two readouts of the same representation, not separate problems.

Generative audio and its evaluation (row 3). The synthetic domain here is produced by modern generators—Stable Audio Open [Evans et al., 2024b] and its predecessor [Evans et al., 2024a], AudioGen [Kreuk et al., 2023], and AudioLDM [Liu et al., 2023]. Their outputs are almost always scored by Fréchet Audio Distance [Kilgour et al., 2019], which compares the distribution of generated audio to real audio and therefore cannot say whether a particular output preserved the event it was conditioned on. That is an open gap for audio-conditioned generation specifically. The sensitive head

Table 2: Corpus composition: clip counts by split (a) and UCS sound morphologies (b).

(a) clip counts by corpus, domain, split

corpus	domain	train	val	test
DCASE-T7	real	4,150	700	700
DCASE-T7	synth	18,165	3,884	3,851
UCS	real	6,636	719	3,065
UCS	twins	6,636	719	3,065

(b) UCS composition by morphology

morphology	events	clips	med. len (s)
transient	11	3,627	3.0
texture	7	2,095	10.4
tonal	4	1,416	5.3
mechanical	5	1,317	11.8
vocal	4	1,141	4.7
ambience	2	473	12.1
whoosh	1	351	0.8
total	34	10,420	5.7

Notes. **(a)** Clip counts by corpus, domain, and split. DCASE-T7 uses DCASE's source-disjoint eval set as the test split; UCS uses FSD50K's train/val/eval splits, keyed on the Freesound uploader so crops of one recording never straddle a boundary. Counts are the 10,420 CLAP-verified real clips retained from 13,579 candidates (77%), each paired with one audio-conditioned synthetic twin, so the real and twin rows match. Instance retrieval queries a synthetic twin against the real gallery, so the 3,065 real test clips form the retrieval gallery used throughout. **(b)** The 34 UCS sound events grouped into the seven broad morphologies (families of how a sound is physically produced) they span. Per sound event, the corpus holds 93–410 real clips (median 338). med. len is the median over a morphology's events of each event's median clip duration (total = corpus-wide median). Real-clip length (natural, before the fixed 5 s analysis window) runs from 0.3 s to the FSD50K 30 s cap, median 5.7 s and right-skewed (mean 8.6 s): 46% of clips are under 5 s and 33% exceed 10 s. Whoosh and transient events are the shortest; mechanical, texture, and ambience the longest.

gives a paired, clip-level realness score, and the instance protocol measures whether a generator's output stays identifiable with its source—diagnostics FAD does not provide.

Domain generalization and contrastive learning (the methods). I build on domain-adaptation tools—Proxy-A-distance [Ben-David et al., 2010], domain-adversarial training (DANN) [Ganin and Lempitsky, 2015], Deep CORAL (correlation alignment) [Sun and Saenko, 2016], and invariant risk minimization (IRM) [Arjovsky et al., 2019]—and on the contrastive family, from supervised contrastive learning [Khosla et al., 2020] to instance discrimination [Wu et al., 2018] and contrastive pretraining (Chen et al., 2020; Radford et al., 2021). The invariant head pursues the same goal as linear concept erasure—removing one factor (here the real-vs-synthetic rendering domain) from an embedding while preserving another (event identity), as Fan et al. [2025] do to deconfound document embeddings. I find that class-level domain-invariance (supcon plus adversarial alignment) overfits the training taxonomy and fails on unseen sound events, while an instance-level objective generalizes—a domain-generalization phenomenon in the synthetic–real setting that, to my knowledge, has not been reported for audio.

3 The Doppelganger Benchmark

Core corpus (DCASE-T7). The DCASE-2023 Task-7 corpus [Choi et al., 2023] provides a ready-made, event-aligned dataset of real and synthetic audio clips spanning 7 classes. It comprises 5,550 real clips, sourced from FSD50K [Fonseca et al., 2022], UrbanSound8K [Salamon et al., 2014], BBC, and Freesound. For the experiments, I adopt DCASE's source-disjoint dev/eval split as train-val/test for the real clips, with the synthetic clips hashed into the same train/val/test splits (Appendix A). In addition, the corpus includes 25,900 synthetic clips generated by 37 different systems. This setup offers a controlled environment for measuring the clean gap between real and synthetic data, as well as for conducting leave-one-class-out probes. Table 2 panel (a) provides an overview.

UCS corpus. Evaluating generalization requires a dataset with many sound events and instance-level pairs. I annotate real audio using the Universal Category System (UCS), mapping the 200 AudioSet [Gemmeke et al., 2017] labels from FSD50K [Fonseca et al., 2022] onto 34 UCS sound events that cover all sound morphologies: transient, texture, tonal, whoosh, vocal, mechanical, and ambience. I verify every clip using zero-shot CLAP [Wu et al., 2023], retaining a clip only if its label ranks in the top-5 by audio–text cosine similarity, which keeps 10,420 out of 13,579 clips (77%), effectively removing FSD50K's noisiest multi-labels. For each verified anchor, I generate a Stable-Audio-Open audio-init twin [Evans et al., 2024b] conditioned on the real clip, resulting in 10,420 instance pairs. The dataset is split following FSD50K's train/val/eval partitions, keyed by Freesound uploader to ensure that crops from the same recording never cross split boundaries. See Table 2 for summary statistics.

Tasks and metrics. The two tasks differ in what counts as a correct match. In sound-event retrieval, I query with a clip and rank clips from the other domain. A match is correct if it shares the query's sound event. I summarize this with mean average precision (mAP)—how highly the same-event clips rank, from 0 to a perfect 1—and report it against a same-domain control, so the headline domain gap is control minus cross-domain mAP—the retrieval performance lost purely by crossing the synthetic–real boundary. In instance retrieval (UCS), only the query's own twin counts as correct (its exact cross-domain counterpart, sharing an instance id). I report R@1, the fraction of queries whose top-ranked match is the true twin, and mean reciprocal rank (MRR), which also gives partial credit when the twin ranks second or third. The headline instance-retrieval numbers carry bootstrap confidence intervals, and on the UCS corpus I use a k-fold leave-classes-out protocol—train with some sound events withheld, test only on those (Sec. 5.3). Full per-event and per-split statistics and the datasheet are in the appendix.

4 Method

The pipeline consists of four steps: employing a fixed backbone, attaching a head whose objective determines the preserved geometry, training with pair-aware batches, and using an evaluation protocol specifically designed to prevent the headline number from being inflated by a favorable gallery.

Step 1: Frozen backbone. I extract embeddings from each frozen encoder—CLAP (512-d), PANNs (2048-d), and AST (768-d). This approach ensures that every comparison isolates the contribution of the head beyond the pretrained representation. Since the backbone's pretraining is uncontrolled, throughout the experiments, "unseen" always refers to data not seen by the head.

Step 2: Head and objective. The head is a compact MLP ($d \rightarrow 512 \rightarrow 256$ with batch-norm and ReLU) with ℓ_2-normalized output, and the choice of objective is the independent variable. The three contrastive heads share a single loss and differ only in which clips count as positives. For normalized outputs z_i and temperature τ, the supervised contrastive loss over a batch is

$$\mathcal{L} = \sum_i \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^\top z_p/\tau)}{\sum_{a \neq i} \exp(z_i^\top z_a/\tau)},$$

where only the positive set $P(i)$ changes between heads. For the invariant head, $P(i)$ is the set of same-event clips, with cross-domain pairs up-weighted (optionally with DANN, CORAL, or IRM). This pulls same-event clips together so that real and synthetic overlap within each event cluster, closing the retrieval gap (though, as shown later, a linear probe can still read the domain). For the sensitive head, $P(i)$ is the same-domain clips within a sound event. It is the deliberate mirror, organizing the space by rendering so that the real-vs-synthetic direction becomes a fidelity axis. For the instance head, $P(i)$ is simply clip i's own generated twin, so it learns the synthetic-to-real mapping rather than class identity. As a compute-matched control, I also train a plain cross-entropy classifier head that matches the instance head in capacity and schedule but is supervised on class labels. This isolates whether any failure arises from the contrastive form or from the use of labels. In the results these appear as the invariant, sensitive, and instance heads, with the class-supervised objectives labelled class-supcon (the invariant head's contrastive objective) and class-CE (the cross-entropy classifier control).

Step 3: Pair-aware training. Since the instance objective requires both a clip and its twin to be present in the same batch to form a positive pair, I batch by instance id. All heads share the same optimizer, schedule, and capacity, ensuring that any observed differences are attributable solely to the objective: AdamW (learning rate 10^{-3}, weight decay 10^{-4}), batch size 1,024, 50 epochs, and supcon temperature $\tau = 0.1$. Exact configurations, seeds, and per-fold class assignments are released with the code.

Table 3: Frozen-encoder cross-domain retrieval gap and domain separability (both corpora, test split).

encoder	corpus	control	synth→real mAP	gap	PAD	domain-probe
CLAP	DCASE-T7	0.935	0.774	$+0.161$	1.27	0.90
CLAP	UCS	0.456	0.343	$+0.112$	1.51	0.88
PANNs CNN14	DCASE-T7	0.798	0.674	$+0.124$	1.39	0.90
PANNs CNN14	UCS	0.318	0.255	$+0.063$	0.88	0.72

Notes. Frozen encoders on each corpus's test split. control = same-domain sound-event mAP (real query, real gallery, with each query's own clip excluded from its gallery); synth→real mAP = cross-domain sound-event retrieval (synthetic query, real gallery); gap = control $-$ cross, the mAP lost by crossing the domain boundary; PAD = Proxy-A-distance (higher = domains more linearly separable); domain-probe = accuracy of a linear real-vs-synthetic classifier. Absolute mAP is lower on UCS because it has 34 sound events versus DCASE-T7's 7, but the gap and the domain probe persist on both corpora: crossing the synthetic–real boundary costs retrieval accuracy, and a linear probe separates the domains well above chance everywhere (CLAP 0.88–0.90). The gap is genuine and linearly accessible, not specific to one corpus.

Step 4: Evaluation protocol. Sound-event retrieval is scored by mAP against a same-domain control, with the gap defined as control minus cross-domain performance. Instance retrieval is scored by R@1 and MRR against the full real test gallery, which includes all sound events as distractors rather than just a held-out-event pool. The gallery size N and chance level $1/N$ accompany each result. Instance retrieval is directional. The reported direction queries a synthetic clip against a real gallery (synthetic→real), the setting relevant to recovering a real source for a generated sound, and the reverse direction is analogous. Generalization is assessed with a deterministic k-fold leave-classes-out protocol (Sec. 5.3), where each fold trains with certain sound events withheld and evaluates only on those unseen events, reporting bootstrap 95% confidence intervals and per-fold paired margins.

5 Results: Does sound identity survive in embedding space?

5.1 How well do frozen encoders separate event from rendering?

The frozen synthetic–real gap. I investigate whether off-the-shelf encoders exhibit a synthetic–real gap by evaluating frozen CLAP and PANNs [Kong et al., 2020] on both corpora, comparing retrieval within the same domain to retrieval across domains. The results are presented in Table 3. In every case, the synthetic–real distinction is present and linearly accessible. For CLAP, cross-domain sound-event retrieval mAP drops by 0.161 on DCASE-T7 and 0.112 on UCS compared to the same-domain baseline; for PANNs, the drops are 0.124 and 0.063, respectively. A linear domain probe separates real from synthetic at 0.88 to 0.90 for CLAP. While absolute mAP is lower on the 34-event UCS corpus than on the 7-event DCASE-T7, the gap and the domain signal remain evident in both. This indicates that sound identity is present but is partially entangled with the method of rendering.

5.2 What do the supervised heads learn—and where do they break?

The class-supervised invariant head reveals a two-part pattern. It performs well within the training taxonomy but fails as soon as a sound event is held out. These are the first two steps in the paper's argument. The third, discussed in Sec. 5.3, is that instance-pair training addresses this failure.

The invariant and sensitive heads. I train two heads that share the same contrastive framework but flip the label, and evaluate them on each corpus's test set with every event seen in training. This isolates what class-label supervision does to the embedding within the taxonomy. The invariant head brings same-event clips together across the domain boundary, while the sensitive head, its intentional counterpart, separates real from synthetic within each event. Table 4 demonstrates that these heads are geometric mirror images on both corpora. The invariant head reduces the by-domain silhouette to 0.00, resulting in real and synthetic clips overlapping within each event cluster, whereas the sensitive head increases it to 0.75 on DCASE-T7 and 0.85 on UCS. On the smaller 7-event DCASE-T7 corpus, the invariant head nearly eliminates the retrieval gap, reducing it from 0.161 to 0.018. On the 34-event UCS corpus, the gap narrows but does not close, with cross-domain retrieval improving from 0.343 to 0.558. Appendix Fig. A1 visually illustrates this reversal, and Appendix Table A3 confirms that plain

Table 4: Retrieval gap and domain silhouette for the invariant and sensitive heads (within-taxonomy, both corpora).

head	corpus	control	cross	gap	$\mathrm{sil}_{\mathrm{ev}}$	$\mathrm{sil}_{\mathrm{dom}}$
frozen	DCASE-T7	0.935	0.774	0.161	0.18	0.04
frozen	UCS	0.456	0.343	0.112	0.07	0.05
invariant	DCASE-T7	0.988	0.970	0.018	0.75	0.00
invariant	UCS	0.658	0.558	0.100	0.23	0.00
sensitive	DCASE-T7	0.206	0.157	0.049	−0.04	0.75
sensitive	UCS	0.129	0.040	0.089	−0.21	0.85

Notes. Each head evaluated on both corpora's test splits (within-taxonomy: every event seen in training). control = same-domain sound-event mAP (real query, real gallery, each query's own clip excluded); cross = synthetic→real sound-event mAP; gap = control − cross; $\mathrm{sil}_{\mathrm{ev}}$/$\mathrm{sil}_{\mathrm{dom}}$ = silhouette score by event class / by domain (higher = tighter clusters on that factor). The two heads are mirror images on both corpora: the invariant head drives the domain silhouette to 0.00 (real and synthetic overlap) while the sensitive head raises it to 0.75–0.85 (real and synthetic split apart). The invariant head nearly closes the retrieval gap on the small 7-event DCASE-T7 corpus (0.161 to 0.018) and shrinks it on the harder 34-event UCS corpus while lifting cross-domain retrieval from 0.343 to 0.558. Appendix Table A3 shows that plain contrastive supervision does this work, with DANN, CORAL, and IRM adding nothing.

contrastive supervision achieves this effect, with DANN, CORAL, and IRM providing no additional benefit. This demonstrates invariance to rendering, achieved within the training taxonomy.

Leave-one-class-out. Does that within-taxonomy success generalize? I hold a single sound event out of training and evaluate the invariant head on it (a leave-one-out probe, distinct from the grouped five-fold leave-classes-out used for the main dissociation), reported in Appendix Table A4. The held-out event performs worse than the untrained frozen baseline (keyboard drops from 0.69 to 0.43, gunshot from 0.81 to 0.31, rain from 0.75 to 0.30), and including more events does not help, since the mean across all 34 UCS events falls from 0.40 to 0.19. Thus, class-label supervision not only fails to generalize to unseen events but actually impairs performance.

5.3 Does instance correspondence generalize across sound events?

Instance vs. class supervision on unseen events. The central test is whether the correspondence generalizes to sound events not seen during training. I compare three objectives—frozen, class-supervised, and instance-pair—under five-fold leave-classes-out, with results in Table 6 and Fig. 2. (Here, "unseen" refers to the head, as the frozen backbone's pretraining is uncontrolled.) Against the full real test gallery (a gallery of $N = 3{,}065$ real clips, so one true match and 3,064 distractors from all sound events per query, chance R@1 $= 0.0003$), the instance head retrieves the exact real twin with R@1 0.800 ($[0.786, 0.813]$), outperforming the frozen encoder (0.611) in every fold (minimum margin $+0.148$) and class supervision (0.269, which is actually detrimental) in every fold (minimum margin $+0.493$). No objective improves sound-event retrieval on unseen events above the frozen encoder (event-mAP near 0.34). A compute-matched cross-entropy classifier matches class-supcon (0.282), indicating that the failure is due to class label training, not the contrastive approach. Only instance-pair supervision generalizes.

Robustness across encoders. The dissociation could be an artifact of a single encoder. I repeat the leave-classes-out comparison on six frozen encoders from three pretraining families, shown in Table 5: supervised (PANNs, AST [Gong et al., 2021]), audio-text contrastive (CLAP), and self-supervised (BEATs [Chen et al., 2023], M2D [Niizumi et al., 2024], AudioMAE [Huang et al., 2022]). For all six encoders, the instance head outperforms frozen in every fold, with R@1 ranging from 0.73 to 0.99, while class supervision falls well below, from 0.27 to 0.49. Five of the six encoders were not involved in the corpus's CLAP verification, so the dissociation is not a result of the CLAP filter. Sound-event retrieval on unseen events remains low throughout. One important detail is that, for the three self-supervised encoders, whose frozen sound-event structure is initially much weaker (full-gallery mAP 0.11 to 0.17 compared to 0.34 for CLAP), the heads recover a small amount of structure (a gain of 0.01 to 0.05 event-mAP; per-encoder values in the released results), but still

Table 5: Instance, class, and frozen retrieval across six encoders and three pretraining families (unseen sound events).

encoder	pretraining	frozen	class head	instance head
CLAP	audio-text contrastive	0.611 [0.594,0.629]	0.269	**0.800** [0.786,0.813]
PANNs CNN14	supervised (AudioSet)	0.657 [0.640,0.673]	0.316	**0.731** [0.715,0.746]
AST	supervised (AudioSet)	0.815 [0.802,0.828]	0.295	**0.940** [0.931,0.947]
BEATs	self-supervised (AS2M)	0.826 [0.812,0.839]	0.335	**0.966** [0.960,0.972]
M2D	self-supervised (AS2M)	0.895 [0.885,0.906]	0.486	**0.988** [0.984,0.992]
AudioMAE	self-supervised (AS2M)	0.804 [0.791,0.819]	0.452	**0.966** [0.960,0.972]

Notes. Full-gallery instance R@1 on unseen sound events (five-fold leave-classes-out, synthetic query, all-sound event real test gallery, $N = 3{,}065$, chance 0.0003); brackets are 95% bootstrap CIs over queries (class-head CIs are omitted for space and are in the released results). Per encoder, a class-supervised and an instance-pair head are trained per fold with identical capacity and schedule. The instance head beats frozen in every fold of every encoder (minimum per-fold margin $+0.05$), and class supervision falls far below frozen on all six—the dissociation is not specific to one encoder or pretraining style. PANNs, AST, BEATs, M2D, and AudioMAE did not participate in the corpus's CLAP-verification.

Table 6: Instance versus class supervision on unseen sound events (five-fold leave-classes-out, CLAP).

variant (unseen sound events)	event-mAP	instance-R@1 (full gallery)
frozen	0.343	0.611 $[0.594, 0.629]$
class-supcon	0.169	0.269 $[0.254, 0.285]$
class-CE (classifier)	–	0.282 $[0.266, 0.297]$
instance	0.262	$\mathbf{0.800}$ $[\mathbf{0.786}, \mathbf{0.813}]$

Notes. Five-fold leave-classes-out on the UCS corpus (CLAP encoder): each fold withholds a set of sound events from training and is evaluated only on those unseen sound events; rows are training objectives. event-mAP = full-gallery synthetic→real sound-event retrieval (not computed for the classifier control, shown as –); instance-R@1 = retrieving the exact real twin of a synthetic query against the full real test gallery (gallery of $N = 3{,}065$ real clips: one true match and 3,064 distractors of all sound events, chance $= 0.0003$), with 95% bootstrap CIs over queries. The instance objective (positive = a clip and its own twin) reaches R@1 0.800, far above frozen; class-supcon and a compute-matched cross-entropy class-CE classifier both fall below frozen (0.27–0.28), so the failure is class-label supervision, not the contrastive form. No objective beats frozen on event-mAP—sound-event structure transfers for none.

fall far short of frozen CLAP, so the qualitative conclusion is unchanged. The instance mapping transfers almost perfectly, while sound-event structure barely transfers. For the strongest encoders, the frozen twin match is already high—M2D reaches 0.90 untrained—making the collapse under class supervision the more informative aspect of the dissociation. Per-event instance R@1 and unseen-event embedding geometry are provided in Appendix Table A2 and Fig. A2.

Finer taxonomy. The 34 UCS categories are intentionally broad, which raises a valid concern. Is "held-out category" too easy a test when the categories are coarse? Appendix Table A5 addresses this by repeating the leave-classes-out protocol at nearly double the granularity, using 65 sound events defined by each clip's most-specific FSD50K/AudioSet leaf label (for example, doorbell, gunshot, thunder, computer keyboard rather than "doors" or "weather"). These are single-labelled categories with ≥ 50 clips each, totaling 8,699 clips, and the same audio, twins, and embeddings are used, with only the labels and head training changed. The dissociation remains unchanged. On unseen fine sound events, the instance head achieves R@1 0.842 ($[0.826, 0.856]$; gallery $N = 2{,}362$, chance 0.04%), outperforming frozen (0.625) and class supervision (0.330, again below frozen) in every fold, while no objective improves sound-event retrieval (event-mAP frozen 0.406, instance 0.323). The qualitative pattern—instance supervision generalizes, class supervision falls below frozen, and sound-event structure does not transfer for either—holds even more clearly at finer granularity, though absolute values are not directly comparable to the 34-event results because the fine gallery is smaller. Approximately 65 is the practical upper limit here, as FSD50K's finer leaf labels are multi-label and long-tailed, leaving too few single-labelled clips per label to hold out an entire label.

5.4 Operating characteristics: ranking, direction, and deduplication

Appendix Table A6 details the practical behavior of the instance head as a tool, all under the same full-gallery, unseen-event protocol. The ranking extends beyond just the top-1. The instance head achieves R@5 0.945 and MRR 0.865 (compared to frozen at 0.801/0.696 and class supervision at 0.419/0.344), so when the exact twin is not ranked first, it is almost always among the top few. The task is also symmetric in direction. Querying with the real clip against the full synthetic gallery (real→synth, the "which generated clip came from this recording?" direction) yields R@1 0.736 versus 0.547 for frozen, indicating that the correspondence is not an artifact of the direction reported. The dataset-hygiene use case also functions at a fixed operating point. When half the queries have their twin removed from the gallery, requiring the tool to abstain, sweeping a single cosine threshold yields best-F1 0.624 (precision 0.645, recall 0.604) for the instance head, compared to 0.479 for frozen.

How much data does the mapping need? This matters for adapting the recipe to a new corpus and generator. I train the instance head on subsampled pair budgets and evaluate on the full real test gallery with all sound events present in training, shown in Appendix Table A7. R@1 increases steadily from 0.681 at 250 pairs to 0.843 at $4{,}000$, compared to 0.611 for frozen and 0.872 with all $6{,}636$ pairs. A few hundred audio-conditioned twins provide most of the benefit, and around $2{,}000$ pairs already reach the level of the headline unseen-event result, though here no events are held out of training.

5.5 Is the sensitive axis a usable realness score?

Does the sensitive axis work as a realness score? I project held-out clips onto its real-minus-synthetic direction and measure how well that separates real audio from synthetic, shown in Appendix Table A8. It separates the training generator from real audio almost perfectly, with AUC 1.000 for real versus Stable-Audio twins at every operating point and 0.999 against the same model's text-only outputs. However, two clear boundaries appear. First, the axis acts as a detector rather than a graded fidelity measure. Within a generator, its score is nearly flat across operating points and uncorrelated with per-clip twin fidelity (Spearman -0.10). Second, it is generator-specific, just like the instance head. ElevenLabs clips, which are synthetic but from an unseen generator, score close to real (AUC 0.747 against real, and higher than Stable-Audio twins on the axis). Thus, both heads tell a consistent story. The correspondence and the detectable rendering signature are properties of a specific generator family, not of "synthetic audio" as a general class. In practical terms, the sensitive head serves as a per-generator audit tool, trained on the generator in use, rather than as a universal realness score.

5.6 A human annotation baseline

I collected a human annotation baseline to place the model numbers on a human scale. Fifty-two annotators on Prolific (49 analyzed after excluding the three who failed both attention checks), each assigned up to 16 real-vs-twin discrimination trials (selecting the real clip from two) and 24 twin-retrieval trials plus 2 catch trials (771 discrimination and $1{,}129$ retrieval judgments analyzed after exclusions), all using the same clips as the models (annotators heard the central 3.5 s of the 5 s model window). Table 7 shows that humans perform between chance and the models on both tasks.

Real-vs-twin discrimination. When distinguishing a real recording from its own synthetic twin (chance 50%), humans achieve only 71.3% ($[66.5, 76.1]$), making errors on nearly three out of ten pairs, while the sensitive head's real-minus-synthetic axis achieves 100% on the same pairs. This quantifies the premise underlying the dataset-hygiene use case. Human curation would accept a substantial fraction of audio-conditioned twins that the model separates perfectly.

Twin retrieval. In the task of identifying the exact real source of a synthetic clip among six same-event candidates (chance 16.7%), humans reach 82.3% ($[78.2, 85.8]$), well above chance, indicating that the correspondence is perceptually real and that the benchmark's pairing is recoverable by independent listeners (majority vote 83.9%, with $299/300$ trials having ≥ 2 raters). However, frozen CLAP already surpasses human performance (92.0%), and the instance head is near the ceiling (99.0%). The task is difficult for humans but straightforward for the appropriate representation, highlighting the gap exposed by the benchmark.

Table 7: Human and model accuracy on the two annotation tasks.

task	chance	human	model
tell the real clip from its twin (2 choices)	50.0	71.3 [66.5, 76.1]	**100.0**
find the real source (6 choices)	16.7	82.3 [78.2, 85.8]	**92.0 / 99.0**

Notes. Accuracy (%) on the identical trials shown to annotators. $N = 49$ analyzed Prolific annotators (3 of 52 excluded for missing both attention checks), 771 discrimination and 1,129 retrieval judgments; human 95% CIs are bootstrapped over annotators. model: for the two-choice task, the sensitive head's real-vs-synthetic direction picks the clip it scores as more real; for the six-choice task, frozen CLAP (92.0) and the instance head (99.0) each pick the nearest of the same six candidates—a restricted six-choice task, not the full-gallery R@1. Annotators heard the central 3.5 s of the 5 s clip the models embed, so the comparison is close but not exactly matched.

5.7 Is it just near-copies? Two controls

Non-learned dedup baselines. If the twins were merely waveform-level near-duplicates, standard deduplication tools would catch them. I run three non-learned baselines on the same full-gallery task, reported in Appendix Table A9. These methods perform poorly: Chromaprint audio fingerprinting, the industry-standard duplicate detector, achieves R@1 0.066, while log-mel and MFCC spectrogram cosine reach 0.236 and 0.229, respectively, compared to 0.611 for frozen CLAP and 0.800 for the instance head. The correspondence resides in the event structure captured by learned representations, not in simple waveform overlap.

Fidelity sweep. Could the instance head simply be matching near-copies? I vary each twin's fidelity to its source and measure retrieval at every level, shown in Fig. 3. Across the fidelity range, the instance head maintains a substantial advantage over frozen (a margin of 0.43 to 0.52), and performance drops to chance only when the twin is entirely independent of its source. This control is measured in CLAP space and is in-domain; the claim regarding generalization across sound events is supported by the leave-classes-out result above.

5.8 The generator-specificity boundary: a transfer matrix

Rather than infer the boundary, I measure it directly by generating twin sets under four generation conditions beyond the core corpus: two more Stable-Audio operating points (`init_noise` 0.45 and 0.75), a second audio-conditioned family (AudioLDM style-transfer at strength 0.5 [Liu et al., 2023]), and a second text-only generator (the Woosh whoosh-domain diffusion model, evaluated on whoosh-sound event anchors). I then cross these with instance heads trained on each family, as shown in Table 8.

Transfer across operating points and families. Table 8 shows that heads trained only on Stable-Audio twins at noise 0.6 transfer effectively to other Stable-Audio operating points, achieving unseen-event R@1 0.862 and 0.728 at noise 0.45 and 0.75, with a margin of 0.17 to 0.22 over frozen. However, they do not transfer to AudioLDM twins, improving frozen only from 0.061 to 0.085, and the reverse transfer is even less effective. A head trained on AudioLDM pairs actually reduces Stable-Audio retrieval, scoring 0.523 compared to 0.611 for frozen, even though an AudioLDM-specific mapping can be learned, as the AudioLDM head achieves 0.147 on its own twins, more than double frozen. Each head thus encodes the rendering map specific to its own generator. One confound acknowledged in the table is that AudioLDM twins have lower fidelity (mean CLAP cosine to source 0.40, with about 10% degenerate), so family and fidelity are somewhat entangled. The reverse case, where the AudioLDM head lowers Stable-Audio retrieval below frozen, cannot be explained by fidelity alone.

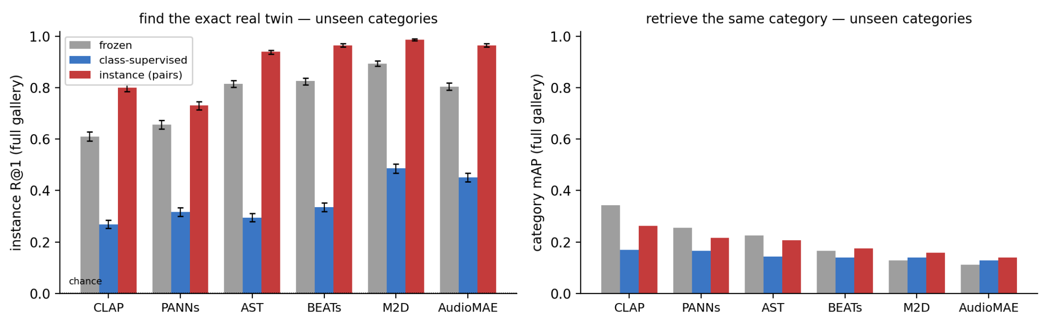
Text-only generators. Both text-only generators fail in the same way in these auxiliary controls. ElevenLabs and the Woosh generator sit near 2–3% R@1, far below the audio-conditioned twins, and no head meaningfully improves them, because the caption bottleneck results in twins that are event-appropriate but not true renderings of the specific clip. The learned mapping is therefore specific to audio-conditioned generation and to the generator family. It serves as an instance-correspondence inverse for a single family, not as a universal synthetic-to-real transformation.

Table 8: Cross-generator transfer matrix (instance R@1).

twin set	conditioning	fidelity	frozen	SAO-trained head	ALDM-trained head
SAO, noise 0.45	audio	0.78	0.696	**0.862**	0.609
SAO, noise 0.6 (core)	audio	0.74	0.611	**0.800**	0.523
SAO, noise 0.75	audio	0.69	0.511	**0.728**	0.434
AudioLDM, strength 0.5	audio	0.40	0.061	0.085	**0.147**
Woosh gen. (whoosh events)	text	0.57	0.023	0.012	0.000
ElevenLabs	text	0.53	0.024	0.027	0.015

Notes. Synthetic→real exact-twin retrieval. fidelity = mean CLAP cosine between each twin and its source. The four audio-conditioned rows use the full real test gallery ($N = 3{,}065$, chance 0.0003); the two text-only rows are auxiliary and not directly comparable, each with its own query set and gallery (Woosh: 86 whoosh-event queries; ElevenLabs: 670 train+val queries against an $N = 7{,}355$ gallery, as no test-split ElevenLabs twins exist—this auxiliary row is not leakage-controlled, since it reuses train/val anchors, yet still fails). SAO-trained head = the paper's instance heads (trained on Stable-Audio noise-0.6 pairs), evaluated under the five-fold unseen-event protocol; ALDM-trained head = an instance head trained on AudioLDM pairs instead. Rows 1–3: the mapping transfers across operating points of the training family. Row 4: it does not transfer across families, in either direction—the ALDM head beats frozen only on its own twins and degrades SAO retrieval below frozen—though ALDM's lower fidelity partially entangles family with fidelity. Rows 5–6: text-only generation breaks instance correspondence for every model and no head recovers it.

Figure 2: Instance, class, and frozen R@1 across six encoders (unseen sound events, full gallery).



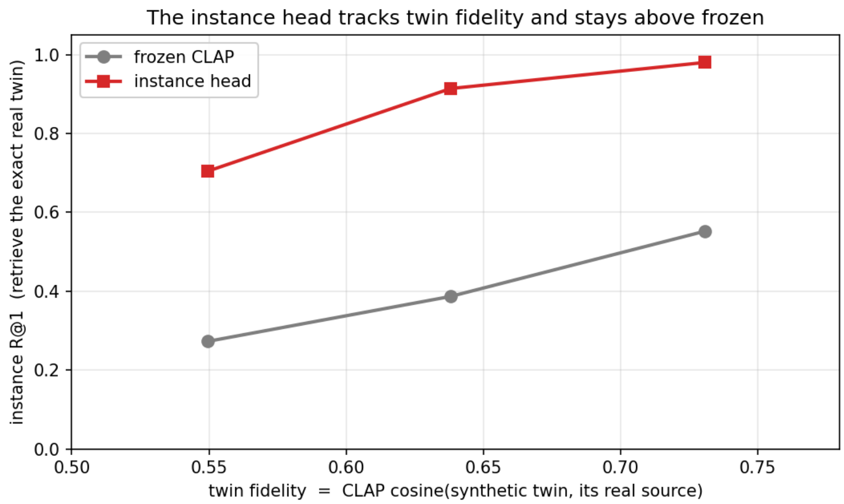
Notes. Held-out (unseen) sound events, across six frozen encoders; bars are full-gallery scores ($N = 3{,}065$), error bars 95% bootstrap CIs. **Left:** synthetic→real instance retrieval—the instance objective (red) lifts R@1 far above frozen (grey) on every encoder, while class supervision (blue) falls far below it. **Right:** sound event retrieval—low for every objective on every encoder; on the three SSL encoders the heads recover a sliver of their weak frozen sound-event structure, never approaching frozen CLAP. Instance matching generalizes across sound events, while category-level retrieval is not improved by any objective.

6 Conclusion

Doppelganger measures whether an audio representation can match a synthetic clip to the specific real recording it was generated from, and whether that ability survives on sound events unseen in training. The answer is a dissociation. Instance correspondence generalizes across sound events while no objective meaningfully improves category-level recognition on unseen ones, and it holds within one generator family but not across families. The broader lesson is conceptual: what the field calls "invariance to rendering" is really two separable capabilities—recognizing the same specific event across the synthetic–real boundary, and recognizing the same kind of event—and a representation can acquire one without the other.

The boundaries this exposes map the next questions. Since each generator family carries its own rendering map, can a head trained on a mixture of families become generator-agnostic, or does adding families merely average incompatible maps? What survives as the twin is generated from text, image, or a physical-parameter description rather than from the source audio? And can a representation made invariant to one rendering process transfer that invariance to another? Each strips away a piece

Figure 3: Instance and frozen R@1 versus twin fidelity (fidelity sweep).



Notes. Instance R@1 against measured twin fidelity—the CLAP cosine between each synthetic twin and its own real source—for frozen CLAP versus the instance head, on in-domain UCS pairs. As twins are rendered less faithfully (moving left) the instance head degrades gracefully and stays above frozen. The head is therefore exploiting a learned correspondence, not near-duplicate copies.

of the conditioning that makes the present correspondence learnable, moving from matching a sound to its twin toward representing the event itself, independent of how it was produced.

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A Datasheet, hosting, and maintenance

Motivation. The dataset measures whether general-purpose audio embeddings represent the identity of a sound event across the synthetic–real boundary, operationalized as cross-domain retrieval at sound event and instance level, with heads that make the representation deliberately invariant or sensitive to the rendering process. It was assembled by the author. The core corpus is a re-framing of an existing public corpus (no new audio collection), and the extensions add generated audio.

Composition (core, DCASE-T7). $31{,}450$ short (≤ 4 s) mono clips: $5{,}550$ real and $25{,}900$ synthetic across 7 event classes (dog_bark, footstep, gunshot, keyboard, moving_motor_vehicle, rain, sneeze_cough), the synthetic clips coming from 37 generator identities (9 Track-A and 27 Track-B DCASE-2023 Task-7 challenge systems plus the baseline). Real audio originates from FSD50K, UrbanSound8K, BBC Sound Effects, and Freesound (provenance columns `orig_dataset`, `orig_id` joined from the DCASE metadata). Splits: DCASE `eval` is the test set (source-disjoint from `dev` by construction). `dev` is split train/val by original Freesound recording id. Synthetic clips are hashed 70/15/15 with every generator present in each split. DCASE generators are event-conditional, so the core corpus has no instance-level pairing.

Composition (UCS corpus). 34 Universal Category System CatIDs mapped from FSD50K's 200 AudioSet labels (`src/taxonomy_ucs.py`), spanning all morphologies. 13,579 real FSD50K clips selected balanced per CatID, then CLAP-verified (kept iff the clip's audio matches its UCS label at top-5 by audio–text cosine), retaining $10{,}420$. One Stable-Audio-Open audio-init twin per verified anchor shares the anchor's `instance_id` and split; 670 ElevenLabs text-only twins (captioned from each clip's Freesound title and tags) support the cross-generator boundary; a fidelity-spectrum subset ($1{,}360$ anchors) is generated at multiple `init_noise` operating points plus a text-only condition. Splits follow FSD50K's train/val/eval, keyed on the Freesound uploader so crops of one recording never straddle a boundary.

Collection and preprocessing. No primary audio collection; the only new human data are the Prolific annotation choices (Appendix B). The pipeline programmatically wraps released archives—walking the directory layout, joining provenance metadata, assigning leakage-safe splits, and emitting one manifest row per clip. Clips are loaded mono, resampled, peak-normalized, and padded or center-cropped to a fixed window so the synthetic–real gap is not confounded by duration or loudness. Embeddings are cached as aligned `(ids, emb)` arrays per encoder. A near-duplicate audit (intra-class cross-split cosine > 0.98) ships with the code.

Uses. Intended: benchmarking audio representations for production-invariant retrieval; training invariance/instance heads; per-generator realness auditing of audio-conditioned generation; dataset hygiene (real/synthetic deduplication). Not recommended: deploying the sensitive head as a general "AI-audio detector"—Sec. 5.5 shows it is generator-specific by construction, and it is provided as a scientific control.

Distribution and licensing. Code is MIT. Real audio is redistributed only where the source license permits (FSD50K, UrbanSound8K, Freesound CC). The 662 BBC Sound Effects clips are research-only and are not redistributed—they ship as manifest rows (`is_cc=0`) with a fetch script. Synthetic DCASE audio is redistributable under the challenge terms. Generated twins ship with the full generation log (event, prompt, seed, steps, cfg, noise level) and are redistributed only when their source clip's license permits derivative outputs; twins of restricted sources ship as manifest rows with the log instead. Each twin is reproducible from the pinned Stable Audio Open model version, environment, and logged seed (approximate rather than bit-identical, since generation is not guaranteed deterministic across hardware and library versions).

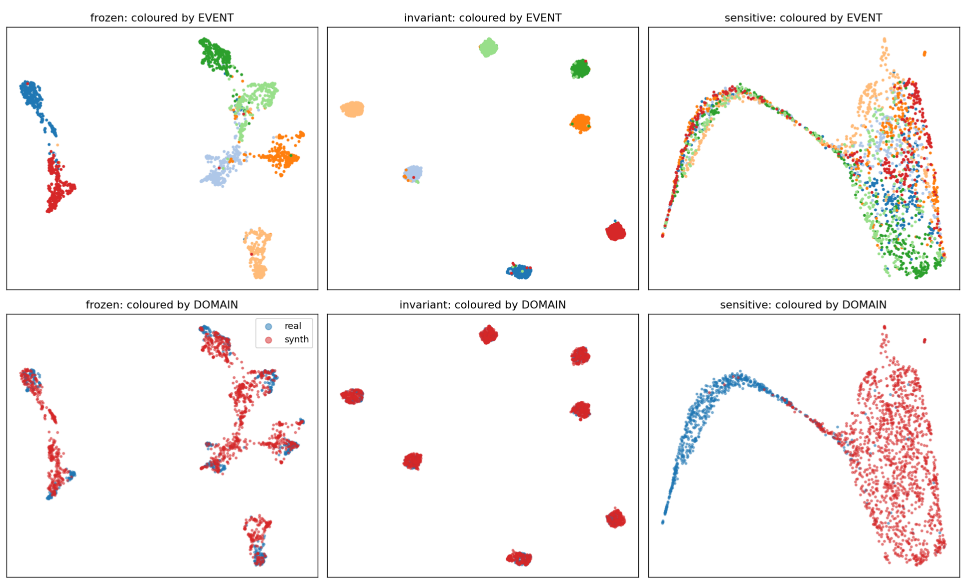
Hosting and maintenance. Code on GitHub (`github.com/elliottash/doppelganger`); audio, manifests, embeddings, and trained heads on the Hugging Face Hub (`huggingface.co/datasets/elliottash/doppelganger`). Manifests are versioned. All derived artifacts (embeddings, heads, results) regenerate deterministically from the manifest and source archives, so corrections are shipped as manifest revisions rather than mutated audio.

B Human annotation baseline: protocol

Annotators ($N = 52$, 49 analyzed) were recruited on Prolific, desktop with working audio required, paid £2.62 for a task of about 12 minutes (well above the platform minimum). The interface presented two blocks: 16 two-alternative forced-choice trials (a real clip and its own synthetic twin in random A/B order, "which is the real recording?") followed by 24 six-way retrieval trials (a synthetic query and six same-event real clips, "which one was this generated from?") interleaved with 2 catch trials whose "synthetic" query was in fact a real recording. All stimuli were the central 3.5 s of the same clips the models embed, loudness- normalized and served under opaque filenames so the answer could not leak through the browser. An annotator who missed both catch trials was excluded from analysis (3 of 52). Everyone who finished was paid regardless. Human 95% CIs are bootstrapped over annotators. Collection stored only the Prolific ID (used solely for compensation and removed from the released, anonymized data), per-trial choices, and response times. Annotators consented on the first screen. Model accuracy in Table 7 is computed on the identical trial sets.

C Within-taxonomy geometry (DCASE-T7)

Figure A1: Embedding geometry of the frozen encoder and the invariant and sensitive heads (DCASE-T7).



Notes. UMAP (Uniform Manifold Approximation and Projection) of the DCASE-T7 test set; columns are the frozen encoder and the two heads. Top row is coloured by event class, bottom row by domain (real vs. synthetic). The invariant head (middle) tightens event clusters while real and synthetic overlap within each—event identity kept, domain removed. The sensitive head (right) does the opposite: it splits real from synthetic and smears the event clusters—domain kept, event removed. Frozen (left) sits between the two. This is the visual form of the silhouette flip in Table 4.

D Per-event analysis

E Supporting result tables

Table A1: Per-event frozen-CLAP cross-domain retrieval (DCASE-T7).

	motor	gunshot	dog_bark	sneeze	rain	footstep	keyboard
mAP	0.852	0.810	0.809	0.793	0.752	0.718	0.692

Notes. Synthetic→real sound event mAP for the frozen CLAP encoder, broken out by DCASE-T7 event class (sorted high to low). Sustained or broadband events (motor, rain) cross the synthetic–real boundary best; fast, granular transient and mechanical events (footstep, keyboard) transfer worst—these are the hardest events to render faithfully and the hardest to retrieve across the gap.

Table A2: Per-sound event instance R@1 on unseen sound events (UCS, CLAP, full gallery).

cat	n	frozen	instance	cat	n	frozen	instance
WOOD	62	0.79	0.95	WHSH	86	0.79	0.83
BODY	80	0.86	0.95	ANML	125	0.65	0.82
DSHS	121	0.81	0.93	BIRD	81	0.52	0.81
VOX	121	0.80	0.93	WIND	26	0.42	0.81
CREA	30	0.70	0.93	CLTH	101	0.63	0.80
CRSH	33	0.85	0.91	BELL	131	0.64	0.78
MUSC	70	0.83	0.89	AMB	17	0.12	0.76
ALRM	95	0.83	0.88	CRWD	98	0.48	0.76
PHON	120	0.82	0.88	CLCK	85	0.60	0.75
FOOT	116	0.70	0.88	TOOL	57	0.56	0.72
KEYS	132	0.76	0.88	COMP	120	0.53	0.71
EXPL	73	0.58	0.88	MECH	62	0.44	0.69
GLAS	129	0.71	0.88	ELEC	40	0.45	0.68
IMPT	69	0.78	0.86	RAIN	99	0.19	0.67
GUN	109	0.58	0.85	VEH	78	0.21	0.50
DOOR	141	0.65	0.85	WATR	121	0.31	0.50
FIRE	134	0.60	0.84	WTHR	103	0.10	0.44

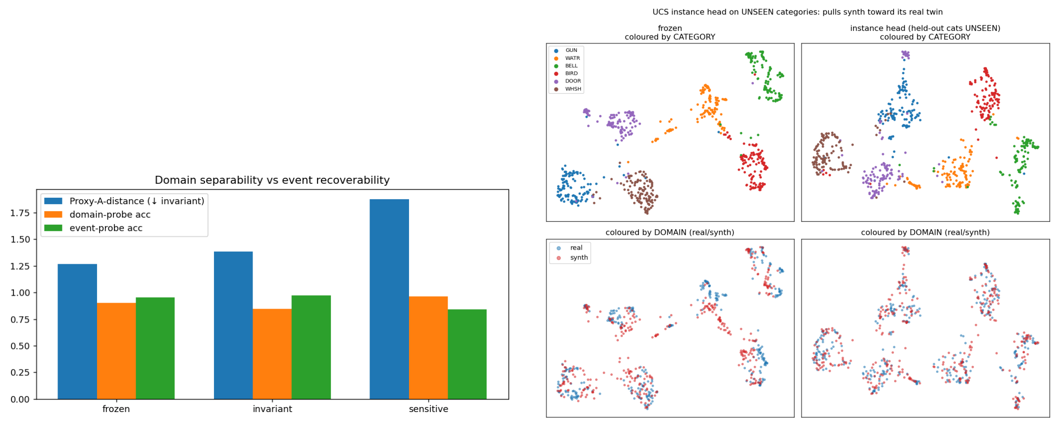
Notes. Exact-twin retrieval (synthetic query, full real test gallery, $N = 3{,}065$) on sound events held out of head training, pooled over the five folds; UCS CatIDs (four-letter Universal Category System codes, e.g. WOOD = wood, DSHS = dishes, WHSH = whoosh; full mapping in the released `src/taxonomy_ucs.py`) sorted by instance-head R@1 (left column high, right column low). n = number of queries per sound event. The instance head improves every sound event over frozen CLAP. The pattern mirrors the DCASE per-event analysis (Table A1): discrete, source-specific events (wood, body, dishes, voice) retrieve best, while sustained textural sound events (weather, water, vehicles, rain)—where distinct recordings genuinely sound alike—remain hardest.

Table A3: Bridging objectives compared: plain contrastive versus DANN, CORAL, and IRM (DCASE-T7).

invariant-head variant	gap	PAD
supcon only	+0.019	1.38
+CORAL	+0.017	1.38
+DANN	+0.019	1.38
+DANN+IRM	+0.018	1.38

Notes. Starting from cross-domain supervised contrast (supcon) on the invariant head, adding CORAL, DANN, or IRM changes the residual gap (control − cross) by less than 0.002 and the Proxy-A-distance (PAD, domain separability) not at all. The contrastive term does the work; the domain-adaptation add-ons are inert here.

Figure A2: Probe separability and the instance head's unseen-event geometry.



(a) **(b)**

Notes. (a) Proxy-A-distance (PAD) and event/domain linear-probe accuracy across the DCASE-T7 heads. The sensitive head sharply raises domain separability (PAD 1.88, domain probe 0.97). The invariant head lowers the domain probe modestly (0.90 to 0.84) and keeps event-probe accuracy high (0.97), but does not erase linear domain information—its PAD (1.39) stays above frozen's (1.27). The invariant head's invariance shows up as the closed retrieval gap and near-zero domain silhouette rather than as domain-probe erasure. (b) UMAP of the instance head on sound events unseen during its training, coloured by domain: real and synthetic mix more thoroughly than under the frozen encoder, consistent with the head having learned a cross-domain instance correspondence rather than a real-vs-synthetic split.

Table A4: Leave-one-class-out retrieval under class supervision versus frozen (DCASE-T7 / UCS).

held-out event	frozen	class-supervised (held out)
keyboard	0.69	0.43
gunshot	0.81	0.31
rain	0.75	0.30
all 34 UCS events (mean)	0.40	0.19

Notes. Cross-domain sound-event mAP on a category the class-supervised head never saw in training. Rows 1–3 are single-event leave-one-out on DCASE-T7; the last row is the mean over the 34 UCS events under leave-classes-out. In every case the class-supervised head scores *below* the untrained frozen encoder on the unseen event—the closed-world failure that the instance objective later repairs.

Table A5: Instance versus class supervision at two taxonomy granularities (65 vs 34 sound events, CLAP, unseen events).

objective	65 fine events ($N = 2{,}362$)		34 events ($N = 3{,}065$)	
	event-mAP	instance-R@1	event-mAP	instance-R@1
frozen	0.406	0.625 [0.606,0.645]	0.343	0.611
class-supcon	0.273	0.330 [0.312,0.349]	0.169	0.269
instance	0.323	**0.842** [0.826,0.856]	0.262	**0.800**

Notes. Five-fold leave-classes-out at two granularities: the 34 UCS events (right) and 65 finer events defined by each clip's most-specific FSD50K/AudioSet leaf label (left; single-labelled, ≥ 50 clips each). Same protocol, audio, twins, and embeddings; only the labels and heads differ. The instance head beats frozen and class-supervision on every fold at both granularities, and no objective improves sound-event retrieval (event-mAP)—the dissociation is not an artifact of coarse categorization. Absolute levels differ because the fine gallery is smaller.

Table A6: Instance-head ranking depth, retrieval direction, and abstention (UCS, unseen events, full gallery).

objective	direction	R@1	R@5	MRR
frozen	synth→real	0.611	0.801	0.696
frozen	real→synth	0.547	0.749	0.642
class-supcon	synth→real	0.269	0.419	0.344
instance	synth→real	**0.800**	**0.945**	**0.865**
instance	real→synth	0.736	0.916	0.816

Notes. Full-gallery retrieval on unseen events ($N = 3{,}065$, chance $\mathrm{R@1} = 0.0003$). R@5 = fraction of queries whose true twin is in the top five; MRR = mean reciprocal rank (partial credit for near misses). The instance head's ranking is deep (R@5 0.945) and works in both directions: querying a real clip against the synthetic gallery (real→synth) gives R@1 0.736. A dataset-hygiene operating point—half the queries have their twin removed, so the tool must also decline—gives best-F1 0.624 (precision 0.645, recall 0.604) for the instance head versus 0.479 for frozen.

Table A7: Instance R@1 versus number of training pairs (subsampled, instance head).

training pairs	0 (frozen)	250	500	1,000	2,000	4,000	6,636 (all)
instance R@1	0.611	0.681	0.722	0.771	0.802	0.843	0.872

Notes. Instance head trained on random subsamples of the audio-conditioned pairs, with all sound events present in both training and test (distinct from the leave-classes-out headline protocol), full gallery $N = 3{,}065$; 95% bootstrap CIs are ± 0.015 throughout. A few hundred pairs already buy most of the effect, and $\sim 2{,}000$ pairs reach the headline unseen-event level—practical for adapting the recipe to a new corpus and generator.

Table A8: Sensitive-axis separability across generators and operating points.

clip source	axis score (higher = realer)	AUC vs. real
real recordings	$+0.63$	—
Stable-Audio (noise 0.6)	-0.63	1.000
Stable-Audio (noise 0.9)	-0.65	1.000
Stable-Audio (noise 1.2)	-0.64	1.000
Stable-Audio (text-only)	-0.56	0.999
ElevenLabs (unseen generator)	$+0.41$	0.747

Notes. Held-out clips projected onto the sensitive head's real-minus-synthetic axis (trained on the UCS train split). AUC uses real as the positive class, so 1.0 means real and synthetic are perfectly separated and 0.5 means indistinguishable. The axis separates real audio from the training generator essentially perfectly (AUC 1.000) at every operating point, but two limits show it is a *detector*, not a fidelity meter: its score is nearly flat across the generator's operating points and uncorrelated with per-clip twin fidelity (Spearman -0.10), and it rates clips from an *unseen* generator (ElevenLabs) as almost real (AUC 0.747 against real) and is scored realer than the training generator's own twins (AUC 0.993 with ElevenLabs treated as the realer class). Realness detection is thus per-generator, mirroring the instance head's boundary.

Table A9: Non-learned deduplication baselines versus learned heads (UCS, unseen events).

method	R@1	R@5	MRR
Chromaprint fingerprint	0.066	0.090	0.079
log-mel spectrogram cosine	0.236	0.318	0.279
MFCC cosine	0.229	0.319	0.275
frozen CLAP	0.611	0.801	0.696
instance head	**0.800**	**0.945**	**0.865**

Notes. Exact-twin retrieval on the identical full-gallery task ($N = 3{,}065$, chance $\mathrm{R@1} = 0.0003$). Chromaprint is the standard audio-fingerprint duplicate detector; log-mel and MFCC cosine are non-learned spectrogram baselines. All fail far below frozen CLAP and the instance head, so the correspondence lives in learned event structure, not fingerprintable waveform overlap.